Venkata Lakshmi Parimala Pasupuleti

Software Engineer · Production Systems · Performance Engineering · Python · CI/CD · Observability · Security · ML Integration · Distributed Systems · Automated Testing

venkatalakshmiparimala@gmail.com | +1-703-261-5410 | linkedin.com/in/venkata-lakshmi-parimala-pasupuleti | github.com/venkatalakshmiparimala | Virginia, USA

PROFESSIONAL SUMMARY

Software Engineer with 9+ years shipping production-grade systems under reliability, security, and performance constraints. At Entrata, owned business-critical backend systems serving 100K+ monthly active users at 99.5% uptime — instrumented with OpenTelemetry, Prometheus, and Grafana, profiled for performance (30% query gain, 50% scalability improvement), and hardened with HIPAA-compliant Libsodium-encrypted pipelines (zero security incidents). Built AgentProbe: open-source AI eval infrastructure with CI-integrated regression harness and benchmarking across multiple model providers. Integrated ML components into production APIs: AI on FHIR (NLP + BERT + embeddings into HIPAA-compliant Flask API); LeaseI (8-agent LangGraph platform with HITL, evaluated at 93.6% precision). M.S. Data Analytics Engineering, George Mason University (GPA 3.87). — github.com/venkatalakshmiparimala/AgentProbe

WORK EXPERIENCE

Blue Horse Digital
Sep 2026 – Present

AI/ML Engineer (Volunteer) · Remote

- Building ML pipelines, data APIs, and reliability tooling; debugging cross-stack issues in data-driven application features.

George Mason University
Aug 2024 – Aug 2025

Research Associate + Teaching Assistant

- Built and validated ML prediction pipelines (Python, PyTorch, HuggingFace) with rigorous quantitative benchmarking (R^2=0.977); automated data integrity checks (Isolation Forest) across 3 research pipelines; statistical validation via hypothesis testing and confidence intervals. Delivered Python programming workshops to 600+ students.

Entrata (Enterprise SaaS / Real Estate Technology)
Sep 2018 – Aug 2024

Senior Software Engineer · Remote

- **Production reliability and performance engineering** across 20+ modules for 100K+ monthly active users at 99.5% uptime — profiled and optimized 200+ SQL queries (30% performance gain via execution plan analysis, indexing, and partitioning); Docker microservices redesign (50% scalability improvement, 40% deployment time reduction); 80% manual operational workload eliminated via Python/Bash automation pipelines.
- **Instrumented production services with observability tooling** (OpenTelemetry, Prometheus, Grafana) — monitoring, alerting, and operational dashboards for mission-critical data pipelines; reduced mean time-to-diagnose by 40%; maintained bug count below 10 for 6+ consecutive months. Owned through production incidents with durable root-cause fixes, not workarounds.
- **Security-hardened production systems** — Libsodium-encrypted pipelines for PHI compliance on AWS S3 migration across hundreds of thousands of resident records; zero security incidents. HIPAA-compliant data isolation across multi-tenant PostgreSQL schemas. Conducted 300+ peer code reviews; mentored 5+ engineers.
- Built fault-tolerant async event-driven systems (RabbitMQ) with retry logic and idempotency guarantees — designed for correctness under concurrent state changes; owned full SDLC across 120+ Agile sprints.

Entrata India, Pune
Jul 2017 – Sep 2018

Associate Software Engineer

- Built backend modules (Java, Python, PostgreSQL); led AWS S3 migration; managed data pipelines across hundreds of thousands of records with output validation and quality gates.

PROJECTS

AgentProbe: AI Evaluation + Benchmarking Infrastructure
CI-INTEGRATED AUTOMATED TESTING

Python · FastAPI · LangGraph · GitHub Actions · OpenTelemetry | github.com/venkatalakshmiparimala/AgentProbe

- 15-endpoint FastAPI backend; 5-metric benchmarking engine; CI-integrated regression harness with golden datasets and reproducible test conditions; distributed model routing; trace-level observability. Built to catch regressions automatically before production - same automated testing and benchmarking discipline this role requires.

AI on FHIR: ML Integration into HIPAA-Compliant Medical Records API
HIPAA / SECURITY ML INTEGRATION

Python · Flask · BERT · Embeddings · FHIR API · React · Docker | github.com/venkatalakshmiparimala/AI_on_FHIR

- Integrated NLP (BERT + embeddings + LLMs) into a production HIPAA-compliant API translating natural-language queries into FHIR API calls — enforced RBAC and PHI access controls across structured medical records; reduced manual retrieval time ~60%.

EDUCATION

M.S. Data Analytics Engineering George Mason University, Virginia — May 2026

GPA: 3.87

B.Tech Computer Science & Engineering JNTU Anantapur — 2017

SKILLS

Languages & Frameworks: Python (8+ yrs), Java (Spring Boot), JavaScript, TypeScript, SQL, PHP, Bash, R — FastAPI, Django, React.js, REST APIs

Systems & Performance: Distributed Systems, Concurrency, Memory Management, Performance Engineering, Performance Profiling, Fault-Tolerant Design, Microservices, Multi-Tenant Architecture, Idempotency, Event-Driven Architecture (RabbitMQ, Kafka), PostgreSQL, Redis, Docker, Kubernetes

Testing, Observability & Security: CI/CD (GitHub Actions), Automated Testing, Benchmarking, TDD, OpenTelemetry, Prometheus, Grafana, Instrumentation, HIPAA, PHI Compliance, Libsodium, SOC 2, NIST, Audit Logging, Terraform

ML & AI Integration: Python, PyTorch, HuggingFace Transformers, scikit-learn, Computer Vision (OCR, BERT embeddings), LangGraph, FastAPI, Model Deployment, LLM Evaluation, AWS (S3, EC2, RDS, Lambda, Bedrock, SageMaker)